

# Mathematics

## Year 4

Cambridge Early Years

Student Manual

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## Both Ways from Zero

### Mathematics · Year 4 · Student Book

Prime Books, the publishing imprint of Prime School, Portugal.

First edition, 2026. Printed in Portugal.

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**Editorial and production.** Written and edited by the Prime School Mathematics Department. Series design, typesetting and illustration direction by Prime Books. Reviewed in manuscript by the school's Year 4 teaching team, whose corrections are built into the structure of every unit.

**Typeset in** Grandstander (headings), Lexend (text), Kalam (annotations) and JetBrains Mono (figures). Lexend was chosen because it is designed to reduce visual crowding for developing readers, and JetBrains Mono because its figures are of equal width, which is what allows a digit to sit squarely in its place-value column. All four are used under the SIL Open Font Licence, version 1.1.

**A note on the numbers.** Every calculation, sequence, place value decomposition, temperature reading and marking total printed in this book was generated and checked by computer during production, and the answer section is derived from the same source as the questions. If you find an error nonetheless, the Mathematics Department would be glad to hear from you.

**Illustrations.** Original artwork commissioned for this edition. Diagrams, number lines, place value charts, thermometers, sequence chains and tables are drawn as vector artwork by the Prime Books typesetting system, so they remain sharp at any size.

Prime School · Rua Poeta Bocage, Quinta do Anjo, Setúbal, Portugal · [www.primeschool.pt](http://www.primeschool.pt)

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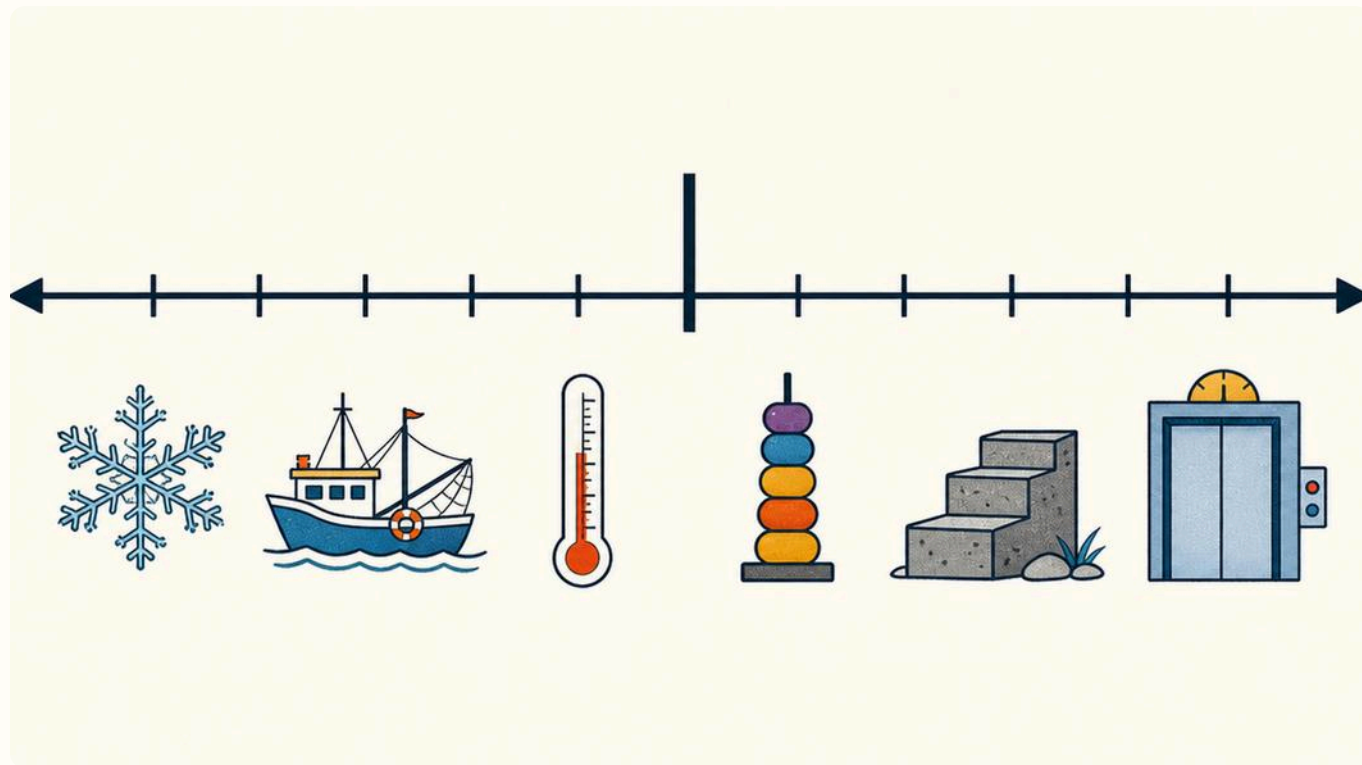
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## WELCOME

# Welcome to Year 4 mathematics

This year you will find out that numbers go further than you think, in both directions.



## THE ONE BIG IDEA

Every number has a **place** and a **direction**. Its place tells you what it is worth. Its direction tells you which side of zero it sits on, and how far.

## WELCOME · CONTINUED

Last year you counted, added, took away and shared. You worked with numbers up to a thousand, and you always started at zero and went upwards. This year the ground opens up underneath zero.

You will meet temperatures in the mountains that fall below zero on a winter morning. You will meet a car park two levels under a building. You will meet a reservoir whose water level is measured downwards from a mark on a wall. In every case you need numbers that are less than nothing, and a way of saying how far below zero they are.

You will also look much more closely at what a digit is worth. In 4 062 the 4 is not worth four. It is worth four thousand, because of the column it sits in. Move that same 4 one column to the right and it is worth only four hundred. Nothing about the digit changed: its place changed.

Those two ideas, place and direction, are what Unit 1 is about, and they come back in every unit after it.

 **Talk about it**

Before you read on, talk to the person next to you. Where have you seen a number with a minus sign in front of it, outside a maths lesson? On a fridge, a weather forecast, a lift button, a news report? Collect three real places between you.

## HOW THIS BOOK WORKS

# Every panel, and what to do with it

Each part of this book is built from the same set of panels. Once you know these nine, you know your way around any page. Six more follow overleaf, with the three drawings that do the explaining.



## ◎ Step onto the line

The opening panel of a part. Read it first: it tells you what the part is about in two or three sentences.

### 0 Step onto the line

Numbers do not stop at zero. In this part you follow the line in both directions.

## ▮ In this part you will

The list of what you will be able to do by the end. Come back to it when you finish, and check yourself against it.

### ▮ In this part you will

- ✓ count on and back in steps of 25
- ✓ read a temperature below zero

## ♀ Out in the real world

A true situation from outside the classroom. It shows you why the maths on the page is worth learning.

### ♀ Out in the real world

*On 8 January the thermometer at the Torre read  $-6^{\circ}\text{C}$ .*

In Faro on the same morning it read  $13^{\circ}\text{C}$ .

## ◎ Warm up your maths

A short question before the teaching starts. It wakes up what you already know. Do it in your head if you can.

WARM  
UP

Count on in fives from 35. Say the first six numbers aloud.

## ↺ Do you remember?

Something from Year 3 that you are about to need. If it feels unfamiliar, say so before the class moves on.

### ↺ Do you remember?

In 583 the 5 stands for five hundred, because it sits in the hundreds column.

## ▮ The material you need

The equipment for a practical task, listed before you start, so nothing stops you halfway through.

### ▮ The material you need

- ☐ a ruler
- ☐ squared paper
- ☐ a red pencil

# Learning, and getting it right

## Learn it

The teaching itself. Read it slowly. If a sentence does not make sense, read it again before you look at the questions.

### Reading below zero

The further left a number sits on the line, the smaller it is. So  $-7$  is smaller than  $-3$ .

## The one big idea

A single sentence worth remembering for the whole year. There are only a few of these in the book.

**A digit's value comes from its column, not from the digit itself.**

## Watch one worked out

A question of the same kind as the ones you are about to do, worked through step by step, with the answer shown. Cover the steps and try it yourself first.

### Watch one worked out

**What is the temperature 6 degrees warmer than  $-4^{\circ}\text{C}$ ?**

1. Start at  $-4$  on the line.
2. Count 6 ticks to the right.

**ANSWER**  $2^{\circ}\text{C}$

## Watch out!

The mistake most people make here, named before you make it. These are the panels to reread the night before a test.

### Watch out!

$-7$  is **colder** than  $-3$ , even though 7 is bigger than 3.

## Your turn

The practice. Work down the list. If you get stuck on one, move on and come back: the questions do not have to be done in order.

### Your turn

- 1 Write the next two terms.  
6, 12, 18, 24, \_\_\_\_, \_\_\_\_
- 2 Which is colder?  
 $-3^{\circ}\text{C}$  or  $-7^{\circ}\text{C}$

## ? Think it through

A question with no quick answer. You are meant to pause here and argue with yourself, or with a partner.

### ? Think it through

Why is the zero in 4 062 doing real work?



# Going further, and the three drawings

## Talk about it

Say your thinking out loud to a partner. Explaining it is how you find out whether you really understand it.

### Talk about it

Is zero a warm number or a cold number?  
Agree an answer between you.

## Number fact

A real fact, checked before it was printed. Every one of these is listed with its source at the back of the book.

### Number fact

The lowest air temperature ever recorded on the Portuguese mainland is  $-16^{\circ}\text{C}$ , at Penhas da Saúde in 1954.

Instituto Português do Mar e da Atmosfera

## One step further

An extra question for when you have finished and want more. It is not harder, it is wider.

### One step further

Find the coldest place in Portugal this week. How far below zero did it go?

## Challenge yourself

A genuinely hard question. You are not expected to get it quickly, and you may need to try something and abandon it.

### Challenge yourself

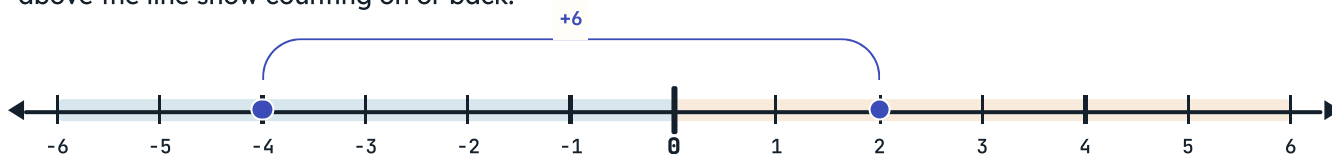
A sequence goes down in steps of 9 and its fourth term is  $-8$ . Where did it start?

## GOING FURTHER, AND THE THREE DRAWINGS · CONTINUED

## The three drawings that do the explaining

## ↔ The number line

Zero is the heaviest tick. Cold and small numbers sit to its left, warm and large numbers to its right. Arcs above the line show counting on or back.



## ▣ The place value chart

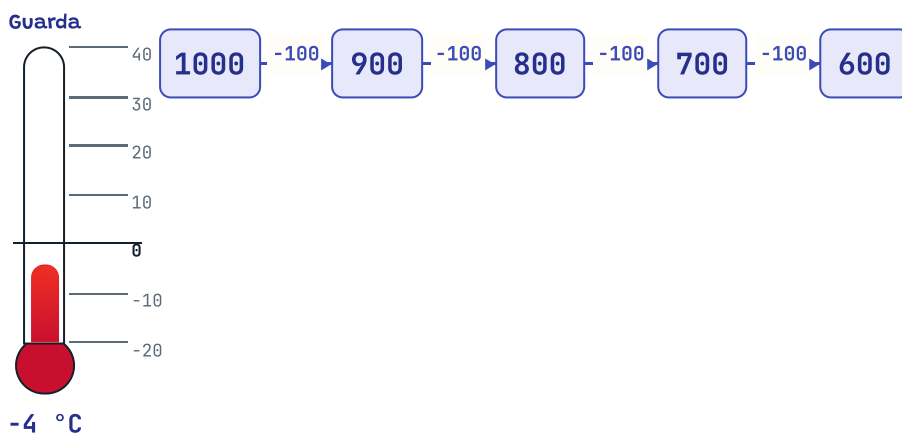
One column for each place. The digits go in the top row, what each digit is worth goes underneath, and the two always agree.

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 4               | 0             | 6         | 2         |
| 4 000           | 0             | 60        | 2         |

$$4\ 062 = 4\ 000 + 60 + 2$$

## = The thermometer and the sequence chain

A thermometer shows one reading on a vertical line. A chain shows a sequence, with the step written on every link.



# How a part ends

Every part finishes the same way, so you always know what is coming: a project to make something with the maths, a page of new questions to test yourself, an honest review, and a list of what you can now do.

## ★ Unit project

A piece of real work that uses everything in the unit. The marking criteria are printed with it, so you know before you start exactly what earns the marks.

### ★ Unit project

## Find zero in six real places

Survey six places where somebody chose a zero, and record what lies on each side of it.

## ▣ Words for the maths wall

Every new word from the part, with its meaning. The same wording appears again in the glossary at the back, so you can look it up later and meet the sentence you were taught.

## ▣ Words for the maths wall

### sequence

a list of numbers in order, following a rule

### term

one of the numbers in a sequence

## ▣ How did it go?

Your own honest review. Three columns: **confident**, **almost there** and **needs more work**. Whatever you tick in the last column is simply what to practise next.

## ▣ How did it go?

Tick one box in each row. Be honest: this page tells you what to practise next.

|                                  | Confident                | Almost there             | Needs more work          |
|----------------------------------|--------------------------|--------------------------|--------------------------|
| count on and back in steps of 25 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## ✓ Now I can

Everything the part set out to teach, in one list. Read it before you start the next part.

### ✓ Now I can

- ☒ count back through zero into negative numbers
- ☒ say what each digit in a number is worth

# Before the first lesson

Five minutes now saves a lot of time later.



## ☐ The material you need

- ☐ an exercise book of squared paper   ☐ two sharpened pencils   ☐ a soft rubber
- ☐ a 30 cm ruler   ☐ a red pencil   ☐ a set of 0 to 9 digit cards



**Picture 1** Everything laid out and ready. The squared paper matters: it is what keeps your columns straight.

## Five things to do now

- 1 Rule a margin two squares wide down the left of every page. Your unit and question numbers go there, and nowhere else.
- 2 Write one digit per square. This is the single habit that prevents most place value mistakes.
- 3 Make your digit cards: cut ten small squares of card and write one digit from 0 to 9 on each. Keep them in an envelope inside the book.
- 4 Start each new part on a fresh page, with the part number at the top.
- 5 Leave four lines at the end of each piece of work. That is where your corrections go.

## Watch out!

Do not rub out a wrong answer until you have worked out **why** it was wrong. A crossed-out mistake with a note beside it is worth more than a clean page.

 **Talk about it**

Agree two class rules for maths this year with your teacher: one about how you ask for help, and one about what you do when you finish early.

How your work will be marked

| What your teacher looks for | What that means in your book                                                                     |
|-----------------------------|--------------------------------------------------------------------------------------------------|
| The right answer            | the number is correct, and the units are written in                                              |
| The working                 | your teacher can see <b>how</b> you got there, not just what you got                             |
| The layout                  | one digit per square, columns lined up, a ruled margin                                           |
| The correction              | a mistake is crossed out once, with the right answer beside it and a note saying what went wrong |

**Table 1** Four things, in the order your teacher reads them. Only the first is about being right.

 **One step further**

When you are stuck, work through these four in order before you put your hand up. Most of the time one of them frees you.

- 1 Read the question again, out loud if you can. Underline the words that tell you what to **do**.
- 2 Write down what you already know from the question. Numbers, units, anything given.
- 3 Draw it. A number line, a chart, a picture of the situation. Almost every question in Unit 1 can be drawn.
- 4 Try a smaller version of the same question with easier numbers, then do the real one the same way.

UNIT 1

# 1

## Numbers and the number system

Counting and sequences · Negative numbers · Understanding place value

### ■ In this unit you will

- ✓ count on and back in tens, hundreds and thousands from any number
- ✓ count back through zero into negative numbers
- ✓ tell a linear sequence from a non-linear one, and describe its rule
- ✓ recognise and extend the pattern made by square numbers
- ✓ read, write and compare numbers that are less than zero
- ✓ explain how negative numbers are used for cold and for depth
- ✓ say what each digit in a four-digit number is worth
- ✓ write numbers up to 10 000 in figures and in words
- ✓ compare and order numbers up to 10 000



# 1 Both ways from zero

## ♀ Out in the real world

*On the morning of 8 January the class rang the weather station at the Torre, the highest point in mainland Portugal.*

The reading was  $-6^{\circ}\text{C}$ . Two hundred kilometres to the south, in Faro, it was  $13^{\circ}\text{C}$  at the same moment. Ana wrote both numbers on the board and asked a question that took the rest of the lesson: **how far apart are they?**

You cannot answer that by taking 6 from 13. The two readings are on opposite sides of zero, so you have to travel up to zero first, and then onwards. That journey is what this unit is about.

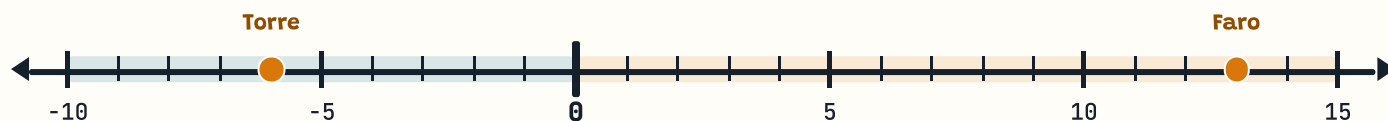


**Picture 1** Ana's class collecting the morning readings. Every number they wrote down that week is used somewhere in this unit.

### THE ONE BIG IDEA

The number line does not stop at zero. It carries on, and the numbers on the far side are called **negative numbers**.

## 1 BOTH WAYS FROM ZERO · CONTINUED



The two readings from 8 January, on one line. Count the ticks between them: from  $-6$  up to  $0$  is  $6$ , and from  $0$  up to  $13$  is  $13$ , so the gap is  $19$  degrees.

🕒 **Number fact**

Portugal's lowest recorded mainland air temperature is  $-16^{\circ}\text{C}$ , measured at Penhas da Saúde in the Serra da Estrela in February 1954. Its highest is  $47.4^{\circ}\text{C}$ , measured at Amareleja in August 2003. That is a spread of  $63.4$  degrees across one small country.

Instituto Português do Mar e da Atmosfera, national climate records



**Part 1** Stepping stones across the Ribeira de Alge. Each stone is a step, and the steps are all the same size. That is exactly what a sequence is.

## 1.1 Counting and sequences

### 0 Step onto the line

Counting in steps is not the same as reciting. When you count on in hundreds from 340 you are adding 100 each time, and you can watch which digit changes and which digits stay still. In this part you will count on and back in tens, hundreds and thousands, cross zero going backwards, and learn to describe a sequence by its rule rather than by its numbers.

#### ■ In this part you will

- ✓ count on and back in steps of 10, 100 and 1000 from any number
- ✓ count back through zero into negative numbers
- ✓ tell a linear sequence from a non-linear one
- ✓ describe a sequence with its term-to-term rule
- ✓ recognise and extend the pattern of the square numbers

WARM  
UP

#### Two ways from 340

Start at 340 and count on in hundreds. Say six numbers aloud. Now start at 340 and count back in hundreds instead. What is the fourth number you say?

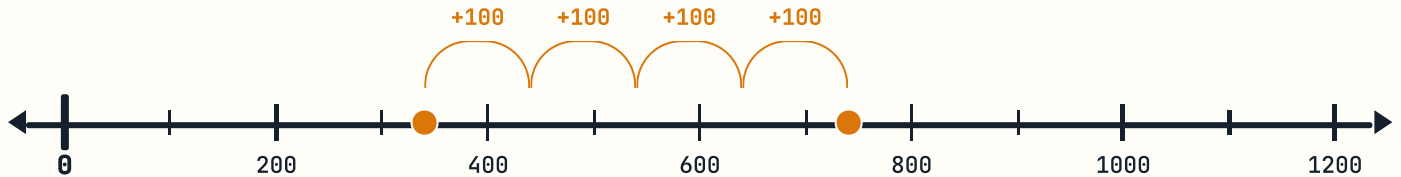
## 1.1 COUNTING AND SEQUENCES · CONTINUED

 Do you remember?

In Year 3 you counted on in tens from 46: 46, 56, 66, 76. The tens digit went up by one each time and the ones digit never moved. Counting on in hundreds works the same way, one column further left.

### Counting on and back in steps

When you count on in **hundreds**, only the hundreds digit changes, until you cross into a new thousand. When you count on in **tens**, only the tens digit changes, until you cross into a new hundred. Watching which column moves is the fastest way to check you have not slipped.

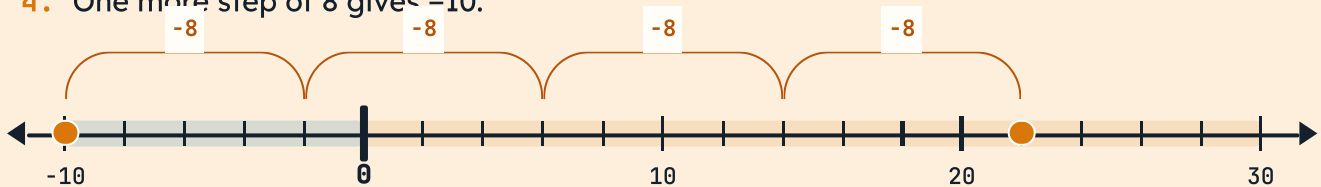


Counting on in hundreds from 340. Each arc is one step of 100, and only the hundreds digit changes.

### Watch one worked out

**Count back in eights from 22. What is the fifth number you say?**

1. Write the starting number: 22.
2. Take away 8 each time: 14, then 6.
3. The next step goes past zero. From 6, take away 8: you go 6 down to 0, then 2 further, so you land on  $-2$ .
4. One more step of 8 gives  $-10$ .



Counting back in eights from 22. Two more steps and you are past zero, on  $-2$  and then  $-10$ .

**ANSWER**  $-10$

### ⚠ Watch out!

When you count back past zero the numbers do not bounce off it. After 6 comes  $-2$ , not 2. Zero is a place on the line, not a wall.

## CONTINUED · CONTINUED

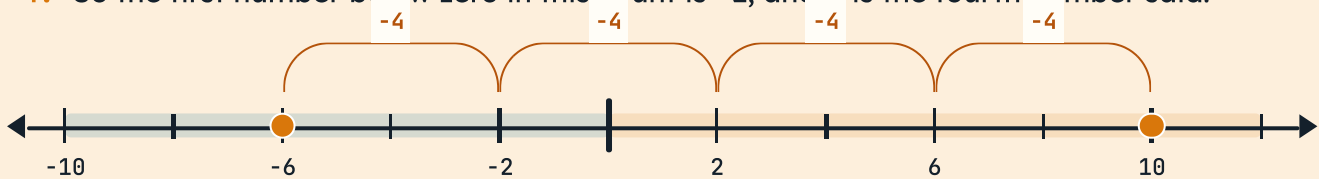
 **Talk about it**

Count back in threes from 10 together, taking turns, and do not stop at zero. Who says  $-8$ ? Now do it again in fours from 10 and see who says it that time.

 **Watch one worked out**

**Count back in fours from 10. Which number in the count is the first one below zero?**

1. Start at 10 and take away 4: that gives 6.
2. Take away 4 again: 2. Still above zero.
3. Take away 4 again. From 2 down to 0 is 2, and 2 further down gives  $-2$ .
4. So the first number below zero in this count is  $-2$ , and it is the fourth number said.



Counting back in fours from 10. The third jump is the one that crosses zero.

**ANSWER** **$-2$ , the fourth number in the count**● **A little more practice**

Three more, to make sure crossing zero feels ordinary.

**A1** Count back in fours from 10. Write five numbers.

**A2** Count back in sevens from 12. Write four numbers.

**A3** Which count reaches  $-6$  exactly?

**a** back in threes from 9

**b** back in fours from 10

**c** back in sixes from 12

**d** back in fives from 9



## ● Your turn

---

Write your answers in your exercise book, one digit per square.

- 1 Count on in hundreds. Write six numbers.

Start at 340.

- 2 Count back in tens. Write five numbers.

Start at 512.

- 3 Count on in thousands. Write four numbers.

Start at 2 604.

- 4 Count back in twenty-fives. Write six numbers.

Start at 75. Do not stop at zero.

- 5 Write the missing numbers in each line.

a 140, 240,  , 440,  , 640      b 1 800, 1 700,  ,  , 1 400

c 18, 12, 6,  ,  , -12      d 2 600, 1 600,  , -400,

- 6 How many steps of 100 from 240 to 940?

steps

- 7 Count back in nines from 15. Write four numbers.

- 8 Start at 8. Count back in fives four times.

Where do you land?

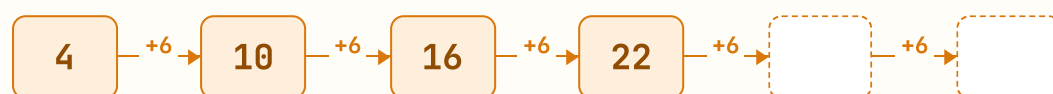
## Sequences and the term-to-term rule

### What makes a sequence linear

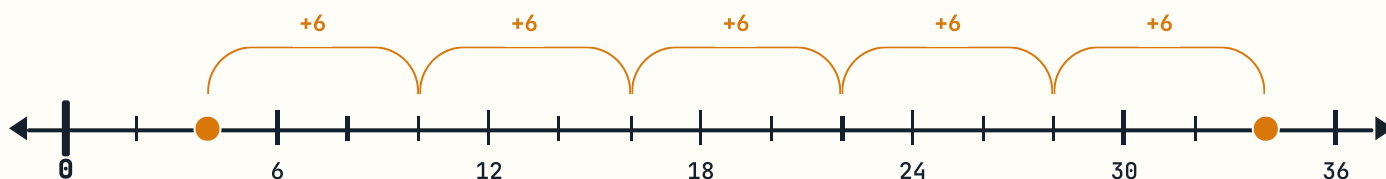
A **sequence** is a list of numbers in order. Each number in it is called a **term**. The **term-to-term rule** tells you how to get from one term to the next.

If the rule adds or subtracts the same amount every time, the sequence is **linear**: the steps are all equal, and on a number line the jumps are all the same width. If the steps change, the sequence is **non-linear**.

Rule: **add 6**



The rule here is **add 6**, so the two blank cards are 28 and 34.



The same sequence on a number line. Every jump is exactly six wide, which is what makes it linear.

Rule: **multiply by 3**



## SEQUENCES AND THE TERM-TO-TERM RULE · CONTINUED

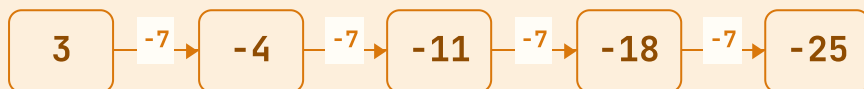
This rule is **multiply by 3**. The steps get bigger and bigger: 2, then 6, then 18, then 54. The jumps are not equal, so this sequence is non-linear.

 **Watch one worked out**

**A sequence starts at 3 and its rule is subtract 7. Write the first five terms.**

1. The first term is given: 3.
2.  $3 - 7$  goes past zero. From 3 down to 0 is 3, and 4 further down gives  $-4$ .
3.  $-4 - 7 = -11$ .
4. Carry on:  $-18$ , then  $-25$ .

Rule: **subtract 7**



**ANSWER** 3, -4, -11, -18, -25

 **Watch out!**

Describing a sequence as **going up in sixes** is not quite a rule. A rule says what to **do** to one term to get the next: **add 6**. Write it as an instruction.

## ● Your turn

9 Write the rule for each sequence.

a 5, 9, 13, 17, 21

b 90, 78, 66, 54

c 2, 4, 8, 16, 32

d 11, 4, -3, -10

10 Write the next two terms in each sequence.

a 7, 15, 23, 31, ,

b 2 400, 2 150, 1 900, ,

c 1, 4, 16, 64, ,

d 14, 8, 2, ,

11 Which of these sequences are linear?

a 6, 12, 18, 24

b 1, 2, 4, 8

c 30, 21, 12, 3

d 2, 3, 5, 8, 12

Write linear or non-linear beside each letter, and say how you decided.

12 A sequence starts at 5. The rule is subtract 8.

Write the first five terms.

13 Find the missing term in the middle.

48, 39, , 21, 12

14 Two sequences share a term. Find it.

The first goes 4, 11, 18, 25 and carries on. The second goes 60, 53, 46, 39 and carries on.

15 Write a linear sequence of five terms that ends at zero.

### ? Think it through

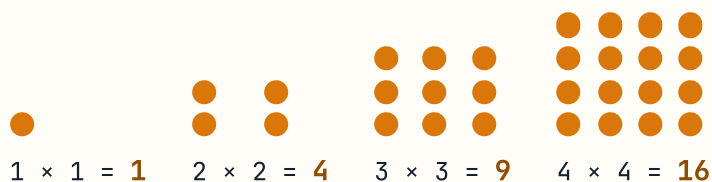
A sequence has the rule **subtract 4** and one of its terms is -6. Could 3 be a term of the same sequence? Try it and explain what you find.

## Square numbers and the patterns they make

### Why they are called square

Take one counter. Now make the smallest square you can with counters: two rows of two, which needs 4. The next square needs three rows of three, which is 9. These numbers are called **square numbers** because that is the shape they make.

You write them as a multiplication of a number by itself:  $3 \times 3 = 9$ . The first six square numbers are 1, 4, 9, 16, 25, 36.



**Picture 2** Azulejo tiles laid out as the first three square numbers. The loose tiles beside them are what you would need to build the fourth.

## SQUARE NUMBERS AND THE PATTERNS THEY MAKE · CONTINUED

## ■ Watch one worked out

The square numbers are 1, 4, 9, 16, 25. What is the gap between each one and the next?

1. Take 1 from 4: the gap is 3.
2. Take 4 from 9: the gap is 5.
3. Take 9 from 16: the gap is 7. The gaps are the odd numbers, in order.
4. So the gap after 25 is 11, and the next square number is  $25 + 11$ .

**ANSWER**

the gaps are 3, 5, 7, 9, 11, and the next square is 36

## → One step further

Add the first few odd numbers and watch what happens:  $1 = 1$ ,  $1 + 3 = 4$ ,  $1 + 3 + 5 = 9$ ,  $1 + 3 + 5 + 7 = 16$ . Every total is a square number. Can you see in the dot pictures above **why** that happens?



## ● Your turn

- 16 Write the first eight square numbers.

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

- 17 Draw the fifth square number as a dot array.  
Use squared paper. How many dots did you draw?

- 18 Which of these are square numbers?

a 20

b 25

c 36

d 40

e 49

f 60

- 19 Write each square number as a multiplication.

a  $16 = \_\_\_ \times \_\_\_$

b  $49 = \_\_\_ \times \_\_\_$

c  $64 = \_\_\_ \times \_\_\_$

d  $81 = \_\_\_ \times \_\_\_$

- 20 Find the square number between 30 and 40.

|  |
|--|
|  |
|--|

- 21 Add the first five odd numbers. What do you notice?

$1 + 3 + 5 + 7 + 9 = \_\_\_\_\_\_$

- 22 How many more counters from the 4th square to the 5th?

|  |
|--|
|  |
|--|

### ▲ Challenge yourself

A square patio is made of square tiles. Marta counts 36 tiles altogether. Her brother says he can make a bigger square using exactly 13 more tiles. Is he right? Draw it on squared paper and explain.

▣ Words for the maths wall

**sequence**

a list of numbers in order, following a rule

**term**

one of the numbers in a sequence

**term-to-term rule**

what you do to one term to get the next

**linear sequence**

a sequence whose steps are all equal

**non-linear sequence**

a sequence whose steps are not all equal

**square number**

the result of multiplying a whole number by itself

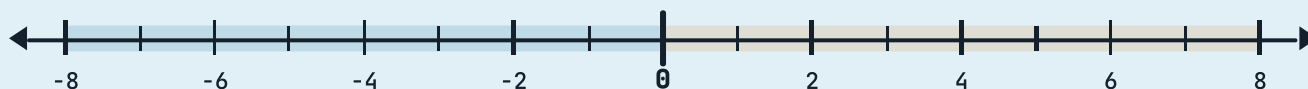


**Part 2** One waterline, two worlds. Above it the numbers go up; below it they go down. Sailors measure both from the same zero.

## 1.2 More on negative numbers

### 0 Step onto the line

A **negative number** is a number less than zero. You write it with a minus sign in front:  $-6$ , and you say it aloud as **negative six**. In this part you will read and write these numbers, place them on a number line, and use them for the two things they are used for most in real life: how cold it is, and how far below a surface something sits.



The line you will be working on for the whole of this part. Notice that it does not stop at zero.

### ■ In this part you will

- ✓ read and write numbers less than zero
- ✓ say a negative number aloud correctly
- ✓ place negative numbers on a number line
- ✓ use negative numbers for temperature and for depth
- ✓ compare two negative numbers and say which is smaller
- ✓ put a set of temperatures in order from the coldest to the warmest

## 1.2 MORE ON NEGATIVE NUMBERS · CONTINUED

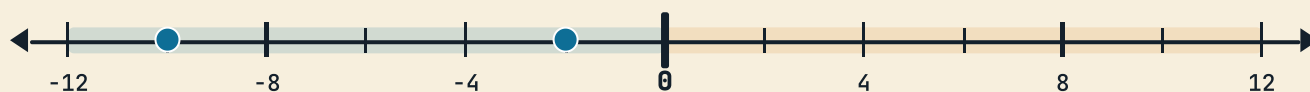
WARM  
UP

## Freezer or fridge

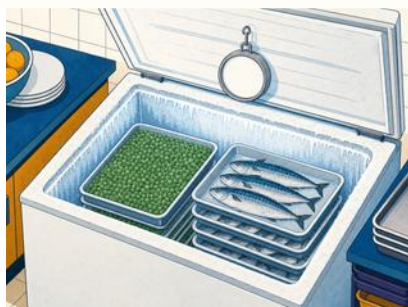
The freezer in the school kitchen is set to  $-18^{\circ}\text{C}$ . The fridge beside it is at  $4^{\circ}\text{C}$ . Which is colder, and how do you know without doing any arithmetic?

## Do you remember?

On the number line in Part 1 you counted back past zero and landed on  $-2$  and  $-10$ . Those were negative numbers. You have already used them.



The two numbers you landed on in Part 1.



**Picture 3** The school kitchen freezer. Its dial reads well below zero all year round.

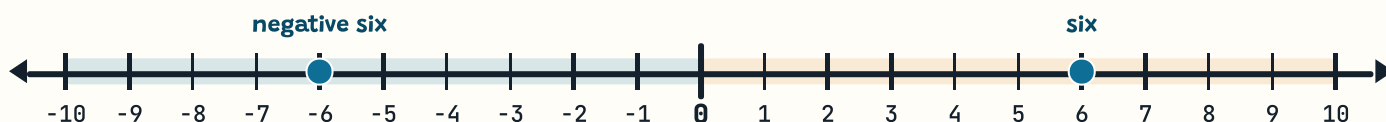


**Picture 4** Emperor penguins breed through an Antarctic winter, where the air can fall to  $-40^{\circ}\text{C}$  and lower.

### Reading and writing numbers below zero

Every number below zero has a matching number above it, the same distance away.  $-6$  is six steps below zero, just as  $6$  is six steps above it. Zero itself is neither positive nor negative: it is the place they are both measured from.

Read the minus sign as the word **negative**.  $-6$  is negative six. Some people say *minus six*, and you will hear it on the weather forecast, but **negative six** is clearer, because minus is also the name of an operation.



Negative six and six are the same distance from zero, one on each side. Count the ticks and check.

#### THE ONE BIG IDEA

The further left a number sits on the line, the **smaller** it is. That is true on both sides of zero.

#### Watch one worked out

Which is smaller,  $-3$  or  $-7$ ?

1. Find both numbers on the number line above.
2.  $-7$  is seven steps to the left of zero.  $-3$  is only three steps to the left.
3. The one further left is the smaller, and that is  $-7$ .

**ANSWER**  $-7$  is smaller

#### ⚠ Watch out!

This is the mistake almost everybody makes once:  $7$  is bigger than  $3$ , so it feels as though  $-7$  should be bigger than  $-3$ . It is not. Adding a minus sign turns the order upside down. Look at the **side** of zero first, then at the **distance**.

## • Your turn

1 Write each number in figures.

a negative four

b negative eleven

c negative twenty

d negative nine

2 Write each number in words.

a -2

b -8

c -15

d -30

3 Which number in each pair is smaller?

a -3 or -7

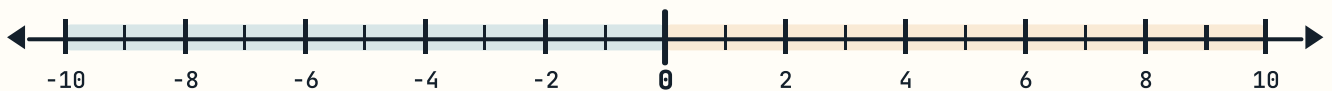
b -1 or 2

c -12 or -9

d 0 or -4

4 Copy the line and mark these four numbers on it.

-8, -3, 2, 7



5 How many steps from the first number to the second?

a -4 to 0

b 0 to -6

c -3 to 5

d -9 to -2

6 Write the number that is three less than each one.

a 2

b 0

c -1

d -8

7 Write the number halfway between each pair.

a -6 and 0

b -4 and 4

c -10 and -2

d -5 and 1

8 True or false? Give a reason for each.

a -5 is less than -2

b 0 is a negative number

c -8 is eight steps below zero

d -1 is bigger than 1

### Talk about it

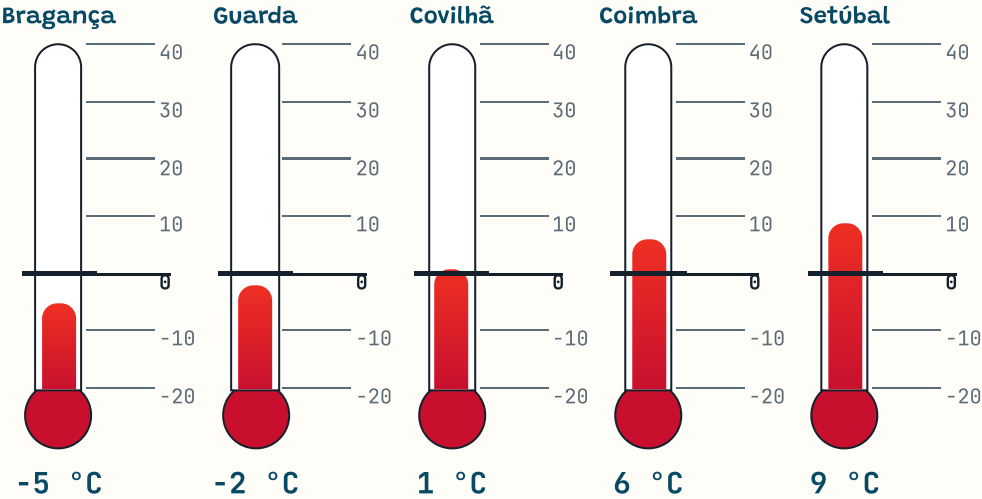
Take turns to say a negative number aloud while your partner writes it down. Swap after four. Did any of them come out wrong? Which ones are easiest to mishear?

## Negative numbers in the real world

Two jobs negative numbers do

**Temperature.** Water freezes at 0 °C. That is why zero sits where it does on a thermometer: it is not the coldest possible temperature, it is the temperature at which water turns to ice. Colder than that and the readings go below zero.

**Depth and level.** Anything measured from a surface can go below it. Sea level is the zero for height on land. The ground floor is the zero for a building. A full mark on a dam wall is the zero for a reservoir.



**Picture 5** Five thermometers, one scale, five readings taken at eight in the morning on the same January day. The mercury height in each is drawn from its reading, so you can compare them by eye.



**Picture 6** A lift shaft. The ground floor is zero, so the car park levels below it are -1 and -2.

|                                                                                         |                              |
|-----------------------------------------------------------------------------------------|------------------------------|
| 2                                                                                       | Floor 2                      |
| 1                                                                                       | Floor 1                      |
| 0                                                                                       | Ground floor                 |
| -1                                                                                      | Level -1                     |
| -2                                                                                      | Level -2 <i>you are here</i> |
| This is why a lift button can say -2. It means two floors below the one you walk in on. |                              |



### 📍 Out in the real world

*The Alge reservoir was 12 metres below its full mark in September, the lowest the village had seen.*

The dam wall carries a measuring post. The full mark is zero. When the water is 12 metres below it, the reading is  $-12$  metres. In a wet March the same post read  $-1$  metres, almost full.



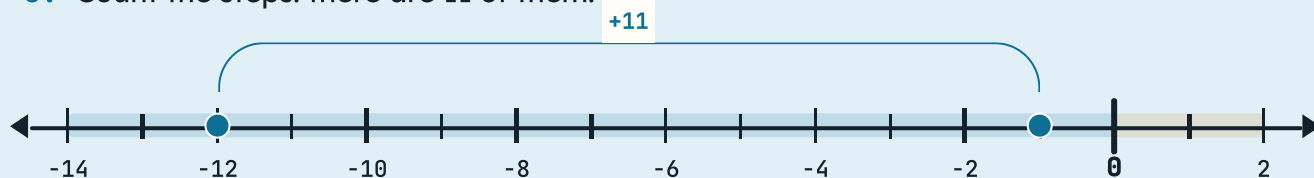
**Picture 7** The measuring post on the Alge dam. Zero is the full mark near the top, so every real reading is negative.

## CONTINUED · CONTINUED

 Watch one worked out

In September the post read  $-12$  metres. By March it read  $-1$  metres. How much did the water rise?

1. Both readings are below the full mark, so both are negative.
2. From  $-12$  up to  $-1$  you count upwards:  $-11$ ,  $-10$ , and so on.
3. Count the steps: there are 11 of them.



**ANSWER** 11 metres

## → One step further

A diver goes down to  $-18$  metres. A gull flies at 25 metres above the same water. How far apart are they? Draw one vertical line with zero at the water surface and mark them both.

## Comparing and ordering temperatures

### Putting readings in order

To compare two temperatures, find them both on the same line. The one further to the left is the colder. To put a whole set in order, work from the leftmost reading and move right.

Always say **which way round** you are ordering. **From the coldest to the warmest** means the smallest number comes first. **From the warmest to the coldest** means the largest comes first. On its own, the word *order* does not tell anybody which you want.

-3 > -7    -1 < 2    -12 < -9    0 > -4

### Watch one worked out

Put these four readings in order, from the coldest to the warmest:  $-3^{\circ}\text{C}$ ,  $2^{\circ}\text{C}$ ,  $-7^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ .

1. Mark all four on one number line.
2. The leftmost is  $-7^{\circ}\text{C}$ , so it is the coldest.
3. Read the rest off the line as you move right:  $-3^{\circ}\text{C}$ , then  $0^{\circ}\text{C}$ , then  $2^{\circ}\text{C}$ .



ANSWER

$-7^{\circ}\text{C}$ ,  $-3^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ ,  $2^{\circ}\text{C}$

Coldest to the warmest

-3    2    -7    0

Write the four readings in the boxes, starting with the coldest.

| Day       | Coldest in the night | Warmest in the day |
|-----------|----------------------|--------------------|
| Monday    | −4 °C                | 5 °C               |
| Tuesday   | −7 °C                | 2 °C               |
| Wednesday | −2 °C                | 9 °C               |
| Thursday  | 1 °C                 | 11 °C              |
| Friday    | −5 °C                | 3 °C               |
| Saturday  | 0 °C                 | 7 °C               |
| Sunday    | −3 °C                | 4 °C               |

**Table 1** Ana’s class recorded these readings at the school weather station over one week in January. Questions 9 to 16 all use this table.

### • Your turn

9 Which night was the coldest?

10 Which day was the warmest?

11 Put the night readings in order, coldest to the warmest.

12 On how many nights did it drop below zero?

nights

13 How much warmer was Monday's day than Monday's night?

degrees

14 Which day had the biggest change from night to day?

15 Work out the gap between the coldest and warmest readings.

Use the whole week. degrees

16 Find two nights exactly five degrees apart.

Coldest to the warmest

-4 -7 -2 1 -5 0 -3



The seven night-time readings from Ana's week.

## • Your turn

- 17 A morning at  $-4^{\circ}\text{C}$  rises 6 degrees. What is it now?

\_\_\_\_\_

- 18 An evening at  $3^{\circ}\text{C}$  falls 9 degrees. What is it now?

\_\_\_\_\_

- 19 Work each one out. Use a number line if it helps.

a  $-5 + 8$

b  $3 - 9$

c  $-2 + 2$

d  $-10 + 4$

- 20 A lift starts at level  $-2$  and rises 5 floors.

Which floor does it reach? \_\_\_\_\_

- 21 A diver at  $-9$  metres rises 4 metres.

What is her depth now? \_\_\_\_\_

- 22 Fill in the missing readings on this thermometer scale.



Copy the line and label every tick.

- 23 Write a temperature colder than  $-8^{\circ}\text{C}$ .

\_\_\_\_\_

- 24 The gap between two readings is 7 degrees. One is  $-4^{\circ}\text{C}$ .

Give both possible answers for the other. \_\_\_\_\_

### ▲ Challenge yourself

A weather station recorded a fall of 3 degrees every hour from midnight. At three in the morning it read  $-4^{\circ}\text{C}$ . What did it read at midnight, and when did it first go below zero?

Words for the maths wall

|                                                                                |                                                                                           |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>negative number</b><br>a number that is less than zero                      | <b>positive number</b><br>a number that is greater than zero                              |
| <b>zero</b><br>neither positive nor negative: the point both are measured from | <b>number line</b><br>a ruled line with equally spaced numbers on it                      |
| <b>sea level</b><br>the zero from which height on land is measured             | <b>degree Celsius</b><br>the unit of temperature, written °C, with water freezing at zero |

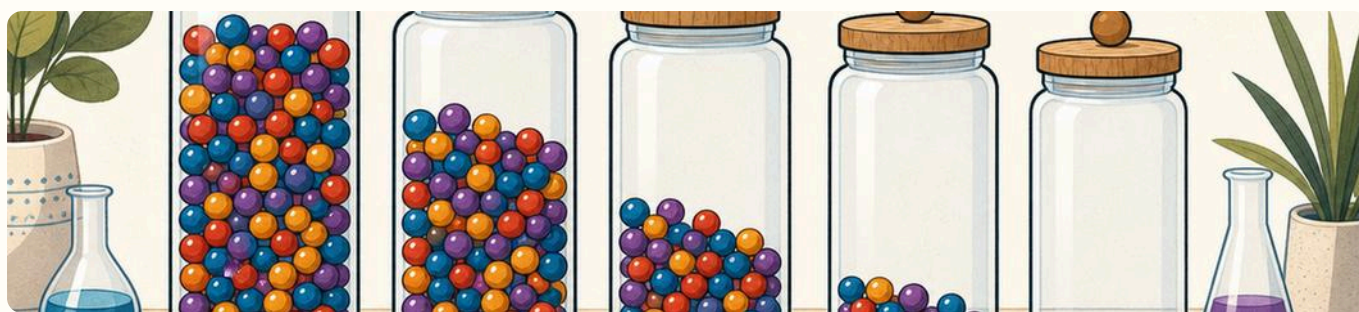
Four places the world counts below zero

| Where              | What zero means there                  | So a negative reading means         |
|--------------------|----------------------------------------|-------------------------------------|
| A thermometer      | the temperature at which water freezes | colder than freezing                |
| A lift             | the floor you walk in on               | a level below the ground floor      |
| A dam or reservoir | the full mark on the wall              | how far the water has dropped       |
| A map of the coast | the surface of the sea                 | how deep the water is at that point |

**Table 2** In every case somebody CHOSE where zero goes. Once it is chosen, both directions have meaning.

**Talk about it**

Between you, find a fifth place where zero has been chosen and readings can fall below it. Say what the zero means there.



**Part 3** Five jars, five different quantities, and the row reads as a staircase. A four-digit number works the same way: each column holds ten times as much as the one on its right.

## 1.3 Understanding place value

### 0 Step onto the line

Look at 4 062. There is a 4, a 0, a 6 and a 2, but the 4 is not worth four. It is worth four thousand, because of the column it stands in. That is what **place value** means: the value of a digit comes from its place.

In this part you will use a place value chart to say what every digit is worth, break a number into its parts and build it back up, write numbers in figures and in words, and put numbers up to 10 000 in order.

#### ■ In this part you will

- ✓ say what each digit in a four-digit number is worth
- ✓ read a place value chart and fill one in
- ✓ break a number into thousands, hundreds, tens and ones
- ✓ build a number back up from its parts
- ✓ write numbers up to 10 000 in figures and in words
- ✓ compare two numbers using the signs  $<$ ,  $>$  and  $=$
- ✓ put numbers up to 10 000 in order, smallest or largest first



## 1.3 UNDERSTANDING PLACE VALUE · CONTINUED

WARM  
UP

## Three digits, three jobs

Write the number 583 in your book. Now circle the digit worth five hundred. Which digit is worth 3, and which is worth 80?

## 🔄 Do you remember?

In Year 3 you worked with three columns: hundreds, tens and ones. This year one more column joins them on the left, the thousands, and everything you already know still works.



**Picture 8** One bead, a stick of ten, a flat square of a hundred, and a solid block of a thousand. Each is ten of the one before it.

 **The place value chart**

A **place value chart** gives every column a name. Reading from the right: **ones, tens, hundreds, thousands**. Each column is worth ten times the one on its right, so as you move left the value multiplies by ten, and as you move right it divides by ten.

The top row of the chart holds the digits of the number. The row underneath holds what each of those digits is **worth**. Add up that second row and you get the number back.

←×10 each step to the left

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 4               | 0             | 6         | 2         |
| 4 000           | 0             | 60        | 2         |

4 062 = 4 000 + 60 + 2

**Picture 9** The place value chart for 4 062. The 4 sits in the thousands column, so it is worth 4 000. The 6 sits in the tens column, so it is worth 60.

THE ONE BIG IDEA

Move a digit one column to the **left** and it is worth **ten times** as much. Move it one column to the **right** and it is worth **a tenth** as much.

What each digit is worth in 4 062

| Digit | Column    | What it is worth                        |
|-------|-----------|-----------------------------------------|
| 4     | Thousands | 4 000                                   |
| 0     | Hundreds  | nothing, but the column is still needed |
| 6     | Tens      | 60                                      |
| 2     | Ones      | 2                                       |

### ■ Watch one worked out

What is the place value of each digit in 5 307?

1. Write the number into a place value chart, one digit per column.
2. The 5 is in the thousands column, so it is worth 5 000.
3. The 3 is in the hundreds column, so it is worth 300.
4. There are no tens, so that column holds a 0.
5. The 7 is in the ones column, so it is worth 7.

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 5               | 3             | 0         | 7         |
| 5 000           | 300           | 0         | 7         |

$$5\ 307 = 5\ 000 + 300 + 7$$

**ANSWER** 5 000 + 300 + 7

### ■ Watch one worked out

Break 9 232 into its parts, then build it back up.

1. Read the digits into their columns: 9 thousands, 2 hundreds, 3 tens and 2 ones.
2. Write what each is worth: 9 000, 200, 30 and 2. Breaking a number up like this is called **decomposing** it.
3. Now add them together to build it back:  $9\ 000 + 200 + 30 + 2$ . Putting it back together is called **composing** it.

9 000

9 thousands

200

2 hundreds

30

3 tens

2

2 ones

**ANSWER**  $9\ 000 + 200 + 30 + 2 = 9\ 232$

## CONTINUED · CONTINUED

**⚠ Watch out!**

Do not throw away a zero. In 4 062 the 0 shows that there are no hundreds. Leave it out and you write 462, which is a completely different number. A zero holds a column open.

**▣ Watch one worked out**

**Which is bigger, 4 062 or 462? Explain how the zero decides it.**

1. Count the digits. 4 062 has four, 462 has three.
2. Put them in a chart. In 4 062 the 4 sits in the thousands column; in 462 the 4 sits in the hundreds column.
3. A digit in the thousands column is worth ten times a digit in the hundreds column, so 4 062 is much bigger.
4. The 0 is what keeps the 4 in the thousands column. Remove it and every digit slides one place to the right.

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 4               | 0             | 6         | 2         |

**ANSWER** 4 062 is bigger, by 3 600

• **Zero holds the column**

Three questions on the one thing that costs the most marks.

**Z1** Write each number in figures. Mind the zeros.

- a three thousand and seven
- b five thousand and ninety

**Z2** Which digit is in the tens column?

- a 4 062
- b 7 305
- c 2 900

**Z3** What happens to 405 if you remove the zero?

Write the new number and say how much smaller it is.

## CONTINUED · CONTINUED

**? Think it through**

Why can you not tell what the digit 6 is worth until somebody shows you the whole number it sits in? Write your answer in one sentence.

## Writing numbers in figures and in words

### The two directions

Sometimes you are given a number in **words** and asked for the **figures**. Sometimes it is the other way round. Both need care, and the zeros are where most marks are lost.

In British English you write **and** before the last part of the number, after the hundreds: **nine thousand two hundred and thirty-two** is 9 232. If there are no hundreds, the **and** comes straight after the thousands: **six thousand and fifty** is 6 050.

### Watch one worked out

Write nine thousand two hundred and thirty-two in figures.

1. Take the parts one at a time. Nine thousand gives a 9 in the thousands column.
2. Two hundred gives a 2 in the hundreds column.
3. Thirty-two gives a 3 in the tens column and a 2 in the ones column.
4. Read the four columns straight off: 9, 2, 3, 2.

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 9               | 2             | 3         | 2         |

**ANSWER** 9 232

## WRITING NUMBERS IN FIGURES AND IN WORDS · CONTINUED

 Watch one worked out

Write six thousand and fifty in figures.

1. Six thousand gives a 6 in the thousands column.
2. There are no hundreds, so put a 0 in the hundreds column to hold it open.
3. Fifty gives a 5 in the tens column, and there are no ones, so another 0 goes at the end.

| Thousands<br>Th | Hundreds<br>H | Tens<br>T | Ones<br>O |
|-----------------|---------------|-----------|-----------|
| 6               | 0             | 5         | 0         |

ANSWER 6 050

 Watch out!

Read the whole thing before you write anything. **four thousand and sixty-two** and **four hundred and sixty-two** start with the same sound but need a different number of columns.

● Your turn

1 Write what each digit is worth in 3 814.

a the 3

b the 8

c the 1

d the 4

2 Copy this chart and fill in 2 705.

| Thousands<br>Th      | Hundreds<br>H        | Tens<br>T            | Ones<br>O            |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

Write one digit in each column, then write what each digit is worth underneath.

3 Write this number in figures.

*seven thousand four hundred and eighteen*

4 Write this number in figures.

*nine thousand and sixty*

5 Write this number in figures.

*four thousand and nine*



- 6 Write this number in words.

**6 285**

---

---

- 7 Write this number in words.

**8 040**

---

---

- 8 Break each number into its parts.

a  $5\,471 =$

b  $3\,208 =$

- 9 Build each number from its parts.

a  $7\,000 + 400 + 30 + 6 =$

b  $2\,000 + 90 + 5 =$

c  $8\,000 + 600 =$

d  $4\,000 + 300 + 1 =$

- 10 Which digit is in the hundreds column?

a 6 294

b 1 738

c 9 005

d 4 160

## Comparing and ordering numbers to 10 000

### Compare from the left, not the right

To compare two numbers, start at the **leftmost** column and work right until the digits differ. The first difference decides it, and you can ignore everything after it.

Two signs do the work.  $<$  means *is less than* and  $>$  means *is greater than*. The pointed end always aims at the smaller number, so  $4\ 026 < 4\ 062$  and  $4\ 620 > 4\ 206$ .

$4\ 026 < 4\ 062$      $4\ 620 > 4\ 206$      $9\ 005 < 9\ 050$

### Watch one worked out

Put these in order, smallest first: 4 062, 4 620, 4 206, 4 026.

1. Every number has a 4 in the thousands column, so that column cannot separate them. Move right.
2. Compare the hundreds: two numbers have 0, one has 2 and one has 6. So the two with 0 hundreds are the smallest pair.
3. Separate that pair on the tens: 4 026 has 2 tens and 4 062 has 6 tens, so 4 026 comes first.
4. Then 4 206, and largest is 4 620.

**ANSWER** 4 026, 4 062, 4 206, 4 620

Smallest first

4062

4620

4206

4026

Four numbers made from the same four digits.

### Watch out!

Within the same number of digits, comparing from the left always works. Line the numbers up one under the other in columns and the answer is usually visible without any arithmetic at all.

## • Your turn

---

11 Copy each pair and write  $<$ ,  $>$  or  $=$  between them.

a 3 714 3 741

b 8 060 8 006

c 5 900 5 900

d 2 199 2 200

12 Put these in order, smallest first.

7 305, 7 035, 7 350, 7 530

13 Put these in order, largest first.

4 880, 4 808, 4 088, 4 800

14 Write the largest number you can with these four digits.

Use 8, 3, 0 and 5 once each.

15 Write the smallest four-digit number with those digits.

Use 8, 3, 0 and 5 once each. Think about the zero.

16 Write a number that fits between the two given.

a 2 400 and 2 500

b 6 099 and 6 110

17 Which number is 100 more than 3 950?

18 Which number is 1 000 less than 4 207?

19 Write these five prices in order, cheapest first.

€1 240, €1 024, €1 402, €1 042, €1 204

20 Make three different four-digit numbers from 9, 4, 0 and 7.

Then put your three in order, largest first.

**→ One step further**

Take the digits 1, 5, 6 and 8. How many different four-digit numbers can you make if each digit is used once? Write them all in order, smallest first, and count them.

**▣ Words for the maths wall****place value**

the value a digit has because of its column

**digit**

one of the ten symbols from 0 to 9

**decompose**

break a number into the parts it is made of

**compose**

build a number up from its parts

**thousands**

the fourth column from the right

**in figures**

written using digits, such as 9 232

**in words**

written out, such as nine thousand two hundred and thirty-two

**is less than**

written with the sign <

# 1 Unit project: the zero line survey

## ★ Unit project

### Find zero in six real places, and measure from it

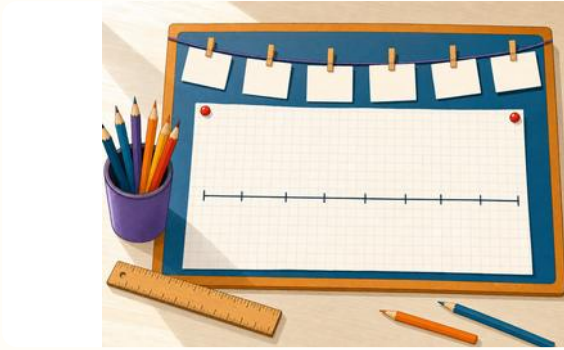
Every measurement in this unit was taken from a zero that somebody chose: the freezing point of water, the full mark on a dam, the ground floor of a building, sea level. Your job is to go and find six of these zeros for yourself, and record what lies on each side of them.

You will work in pairs, over one week, and present your survey on a single sheet of squared paper.

#### ☐ The material you need

- ☐ one sheet of A3 squared paper
- ☐ a 30 cm ruler
- ☐ a sharp pencil and a red pencil
- ☐ a thermometer from the science cupboard
- ☐ the school weather readings for the week

## 1 UNIT PROJECT: THE ZERO LINE SURVEY · CONTINUED



**Picture 10** One pair's finished survey sheet. The line runs the full width of the paper and every reading is pegged to it in the right place.

**What to do**

- 1 Choose six places where a zero has been chosen for a reason. At least one must be a temperature, and at least one must be a depth or a level.
- 2 For each one, write down what the zero **means** in that place, in your own words. This is the part that carries the most marks.
- 3 Take or find one real reading on each side of each zero.
- 4 Rule one long horizontal line across your sheet. Mark zero heavily in the middle, and scale the ticks so your largest and smallest readings both fit.
- 5 Plot all twelve readings on the line, in red above and in pencil below.
- 6 Work out the gap between your coldest and warmest readings, and show your working beside the line.

## What to hand in, and how it will be marked

### What to hand in

- Your survey sheet, with the number line drawn and all twelve readings plotted.
- A short written note for each of the six zeros, saying what it means and who chose it.
- The gap between your coldest and warmest readings, with the working shown.
- One sentence saying which of your six zeros surprised you most.

Hand in the sheet and the notes together. Write both names on the back, not the front, so the survey is what is seen first.

| How your project will be marked                             | Weight      |
|-------------------------------------------------------------|-------------|
| The six zeros are real, and each one is explained clearly   | 30%         |
| Readings are correctly placed on a line with an even scale  | 25%         |
| The gap between coldest and warmest is worked out correctly | 20%         |
| Negative numbers are written correctly throughout           | 15%         |
| The sheet is clear enough for another pair to read          | 10%         |
| <b>Total</b>                                                | <b>100%</b> |

### Talk about it

Before you hand the survey in, swap sheets with another pair. Ask them one question about their scale and answer one about yours. Change anything their question showed you was unclear.

### Challenge yourself

Find a zero that is not a temperature, a depth or a floor level. There are several in everyday life: think about time, money owed, or the score in a game. Add it to your sheet as a seventh zero and explain why it counts.

# Check your progress

Work on your own, and without help, so that the result tells you something true. These questions are new: none of them has appeared earlier in the unit.



- **Twelve questions**

Twelve questions. Give yourself one mark for each part you get right, then turn the page.



## UNIT 1 • CONTINUED

- 1** Count on in nineties from 660. Write five numbers.

- 2** Count back in sixes from 11. Write five numbers.

Keep going past zero.                                                       

- 3** Write the term-to-term rule for this sequence.

26, 34, 42, 50

- 4 Is this sequence linear or non-linear? Say why.

3, 6, 12, 24, 48

- 5 Write the seventh square number.

Page 10 of 10

- 6 Which is colder,  $-9^{\circ}\text{C}$  or  $-6^{\circ}\text{C}$ ?

\_\_\_\_\_

- 7** Put these readings in order, from the coldest to the warmest.

$-9\text{ }^{\circ}\text{C}$ ,  $4\text{ }^{\circ}\text{C}$ ,  $-1\text{ }^{\circ}\text{C}$ ,  $-6\text{ }^{\circ}\text{C}$ ,  $0\text{ }^{\circ}\text{C}$

- 8 A morning at  $-6^{\circ}\text{C}$  rises 11 degrees. What is it now?

Page 10 of 10

- 9** Write what each digit is worth in 5 730.

a the 5

**b** the 7

c the 3

d the 0

- 10** Write this number in figures.

two thousand eight hundred and six

- 11** Write this number in words.

6 408

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- 12** Put these in order, smallest first.

5 730, 5 073, 5 370, 5 807

UNIT 1

# How did it go?

This page is for you, not for a mark. Read each row, decide honestly, and tick one box. Whatever you tick in the third column is what to practise next, and that is useful information, not bad news.



☐ How did it go?

Tick one box in each row. Be honest: this page tells you what to practise next.

Part 1 · Counting and sequences

|                                                                   | Confident                | Almost there             | Needs more work          |
|-------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| count on and back in tens, hundreds and thousands from any number | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| count back through zero into negative numbers                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| tell a linear sequence from a non-linear one                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| describe a sequence using its term-to-term rule                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| recognise and extend the square numbers                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## UNIT 1 · CONTINUED

## Part 2 · More on negative numbers

|                                                              | Confident                | Almost there             | Needs more work          |
|--------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| read and write numbers less than zero                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| explain how negative numbers are used for cold and for depth | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| compare two negative numbers and say which is smaller        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| put temperatures in order from the coldest to the warmest    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## Part 3 · Understanding place value

|                                                     | Confident                | Almost there             | Needs more work          |
|-----------------------------------------------------|--------------------------|--------------------------|--------------------------|
| say what each digit in a four-digit number is worth | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| read a place value chart and fill one in            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| write numbers up to 10 000 in figures and in words  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| compare and order numbers up to 10 000              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

If you ticked **needs more work** on any row, write the part number beside it (1.1, 1.2 or 1.3) and go back to the **Watch out!** panel in that part. It names the exact mistake most people make there.

## UNIT 1

# Now I can

Everything this unit set out to teach, in one place. Read it through before you start Unit 2.



## ✓ Now I can

- ✓ count on and back in steps of ten, a hundred and a thousand, starting from any number
- ✓ count backwards through zero and keep going into the negative numbers
- ✓ tell a linear sequence from a non-linear one by looking at its steps
- ✓ describe any sequence by its term-to-term rule, as an instruction
- ✓ recognise the square numbers and explain the shape they are named for
- ✓ read, write and say aloud numbers that are less than zero
- ✓ explain why a freezer, a lift and a reservoir all need negative numbers
- ✓ compare two negative numbers and say which one is smaller
- ✓ put a set of temperatures in order from the coldest to the warmest
- ✓ say what every digit in a four-digit number is worth, and why
- ✓ break a four-digit number into its parts and build it back up
- ✓ write any number up to 10 000 in figures and in words
- ✓ compare numbers with the signs  $<$ ,  $>$  and  $=$ , and order them

## UNIT 1 · CONTINUED

## CARRY THIS INTO UNIT 2

Place tells you what a digit is worth. Direction tells you which side of zero a number is on. Everything in Unit 1 came from those two ideas, and so does most of Unit 2.



Unit 2 begins in the market, where the numbers are added and taken away rather than counted.

## REFERENCE

# Words for the maths wall

Every word this unit introduced, defined the same way as it was defined on the page where you met it. The part where each word appears is given so you can go back to it.



**compose** build a number up from its parts.  $4\ 000 + 60 + 2$  composes  $4\ 062$ . Part 3

**decompose** break a number into the parts it is made of.  $4\ 062$  decomposes into  $4\ 000 + 60 + 2$ . Part 3

**degree Celsius** the unit of temperature, written  $^{\circ}\text{C}$ , with water freezing at zero. That is why zero sits where it does on a thermometer. Part 2

**digit** one of the ten symbols from 0 to 9. A digit's value depends on the column it stands in. Part 3

**in figures** written using digits, such as  $9\ 232$ . Part 3

**in words** written out, such as nine thousand two hundred and thirty-two. Part 3

**is less than** written with the sign  $<$ . So  $4\ 026 < 4\ 062$ . The pointed end always aims at the smaller number. Part 3

**linear sequence** a sequence whose steps are all equal. For example 4, 10, 16, 22. Part 1

**negative number** a number that is less than zero. It is written with a minus sign in front:  $-6$  is negative six. Part 2

**non-linear sequence** a sequence whose steps are not all equal. For example 1, 3, 9, 27. Part 1

**number line** a ruled line with equally spaced numbers on it. It carries on in both directions past zero. Parts 1 and 2

**place value** the value a digit has because of its column. The 4 in  $4\ 062$  has a place value of four thousand. Part 3

**positive number** a number that is greater than zero. Part 2

**sea level** the zero from which height on land is measured. Below it, heights are negative. Part 2

**sequence** a list of numbers in order, following a rule. Part 1

**square number** the result of multiplying a whole number by itself.  $3 \times 3 = 9$ , so 9 is a square number. Part 1

**term** one of the numbers in a sequence. Part 1

**term-to-term rule** what you do to one term to get the next. Write it as an instruction, such as *add 6*. Part 1

**thousands** the fourth column from the right. Part 3

**zero** neither positive nor negative: the point both are measured from. Parts 1 to 3

## REFERENCE

# Answers

Use these to mark your own work **after** you have finished, never while you are working. Where a question asks you to explain something, the answer here gives the point your explanation should make, not the exact words to write.



## Part 1 · Counting and sequences

- 1** 340, 440, 540, 640, 740, 840
- 2** 512, 502, 492, 482, 472
- 3** 2 604, 3 604, 4 604, 5 604
- 4** 75, 50, 25, 0, -25, -50
- 5** a 340, 540; b 1 600, 1 500; c 0, -6; d 600, -1400
- 6** 7 steps
- 7** 15, 6, -3, -12
- 8** -12
- 9** a add 4; b subtract 12; c multiply by 2; d subtract 7
- 10** a 39, 47; b 1 650, 1 400; c 256, 1 024; d -4, -10
- 11** a linear, steps of 6; b non-linear, the steps double; c linear, steps of -9; d non-linear, the steps grow by one each time
- 12** 5, -3, -11, -19, -27
- 13** 30
- 14** 39, which is the fourth term of the first sequence and the fourth of the second
- 15** many answers, for example 20, 15, 10, 5, 0
- 16** 1, 4, 9, 16, 25, 36, 49, 64
- 17** 25 dots, in five rows of five
- 18** 25, 36 and 49 are square numbers; 20, 40 and 60 are not
- 19** a  $4 \times 4$ ; b  $7 \times 7$ ; c  $8 \times 8$ ; d  $9 \times 9$
- 20** 36
- 21** 25, which is a square number, and the total of the first five odd numbers is always the fifth square number

## REFERENCE · CONTINUED

**22** 9 more counters, because the gaps between square numbers are the odd numbers in order

**Part 2 · More on negative numbers**

**1** a -4; b -11; c -20; d -9

**2** a negative two; b negative eight; c negative fifteen; d negative thirty

**3** a -7; b -1; c -12; d -4

**4** check that -8 is eight ticks left of zero and 7 is seven ticks right of it

**5** a 4 steps; b 6 steps; c 8 steps; d 7 steps

**6** a -1; b -3; c -4; d -11

**7** a -3; b 0; c -6; d -2

**8** a true; b false, zero is neither; c true; d false, -1 is smaller than 1

**9** Tuesday, at -7 °C

**10** Thursday, at 11 °C

**11** -7 °C, -5 °C, -4 °C, -3 °C, -2 °C, 0 °C, 1 °C

**12** 5 nights

**13** 9 degrees warmer

**14** Wednesday, with a change of 11 degrees

**15** 18 degrees, from -7 °C up to 11 °C

**16** Tuesday at -7 °C and Wednesday at -2 °C; Friday at -5 °C and Saturday at 0 °C; Monday at -4 °C and Thursday at 1 °C

**17** 2 °C

**18** -6 °C

**19** a 3; b -6; c 0; d -6

**20** floor 3

**21** -5 metres

**22** the ticks are -15, -10, -5, 0, 5, 10, 15

**23** any reading colder than -8 °C, for example -9 °C

**24** 3 °C or -11 °C

**Part 3 · Understanding place value**

**1** the 3 is worth 3 000; the 8 is worth 800; the 1 is worth 10; the 4 is worth 4

**2** 2, 7, 0, 5, worth 2 000, 700, 0 and 5

**3** 7 418



## REFERENCE · CONTINUED

- 4** 9 060
- 5** 4 009
- 6** six thousand two hundred and eighty-five
- 7** eight thousand and forty
- 8** a  $5\,000 + 400 + 70 + 1$ ; b  $3\,000 + 200 + 8$
- 9** a 7 436; b 2 095; c 8 600; d 4 301
- 10** a 2; b 7; c 0; d 1
- 11** a <; b >; c =; d <
- 12** 7 035, 7 305, 7 350, 7 530
- 13** 4 880, 4 808, 4 800, 4 088
- 14** 8 530
- 15** 3 058, because a four-digit number cannot start with 0
- 16** a any number between them, for example 2 450; b 6 100
- 17** 4 050
- 18** 3 207
- 19** €1 024, €1 042, €1 204, €1 240, €1 402
- 20** many answers; check the three are in order with the largest first

## Check your progress

- 1** 660, 750, 840, 930, 1 020
- 2** 11, 5, -1, -7, -13
- 3** add 8
- 4** non-linear, because each term is double the one before, so the steps keep growing
- 5** 49
- 6**  $-9^{\circ}\text{C}$
- 7**  $-9^{\circ}\text{C}$ ,  $-6^{\circ}\text{C}$ ,  $-1^{\circ}\text{C}$ ,  $0^{\circ}\text{C}$ ,  $4^{\circ}\text{C}$
- 8**  $5^{\circ}\text{C}$
- 9** the 5 is worth 5 000; the 7 is worth 700; the 3 is worth 30; the 0 shows there are no ones
- 10** 2 806
- 11** six thousand four hundred and eight
- 12** 5 073, 5 370, 5 730, 5 807

## REFERENCE

# For teachers

Notes on this unit, collected here rather than printed on the pupils' pages.



## For teachers

**Sequencing.** Part 1 deliberately introduces counting back past zero before Part 2 names negative numbers. Pupils meet the idea as a movement they can perform, and only then meet the vocabulary. If your class is confident, the two parts can be taught in one week; if not, Part 1 stands alone.

**The Watch out! panels are the assessment map.** Each one names a specific misconception: that counting back stops at zero; that  $-7$  is greater than  $-3$  because 7 is greater than 3; that a zero digit can be dropped; that *going up in sixes* is a rule. When pupils tick **needs more work** on the review page, send them to the matching panel rather than to the whole part.

**Part 3 and the syllabus.** The scheme of work document repeats topic 1.2's objectives under the heading 1.3. Part 3 of this book teaches the place value objectives that Year 4 requires: the value of each digit in a four-digit number, composing and decomposing, figures and words, and comparing and ordering to 10 000. If your copy of the scheme still shows the repeated text, this is the reading we have used.

**Practical work.** The Part 2 thermometer readings are real in shape but were generated for this book. Replace them with your own school's weather station data if you have a week of it: the questions are written so that only the numbers change, not the tasks.

**Marking the project.** The published criteria carry thirty marks for explaining the six zeros, which is more than for the plotting. That weighting is deliberate. A pupil who can say why the full mark on a dam wall is a sensible zero has understood the unit; a pupil who plots beautifully without that has not.

**Answers.** The answer section was generated from the same source as the questions, so it is arithmetically reliable. Where a question asks for an explanation, the answer gives the point to look for, not a form of words to require.

This page may be photocopied for use with your own classes.

## REFERENCE

# Sources and references

Every fact printed in a **Number fact** panel, and every real-world convention used in a scenario, comes from one of these.



- § Instituto Português do Mar e da Atmosfera. National climate normals and extreme value records for mainland Portugal, including the mainland minimum of  $-16^{\circ}\text{C}$  at Penhas da Saúde, 1954, and the maximum of  $47.4^{\circ}\text{C}$  at Amareleja, 2003. [www.ipma.pt](http://www.ipma.pt)
- § Agência Portuguesa do Ambiente. Reservoir storage bulletins, which report levels as a depth below the full storage mark; the source of the convention used in the Alge reservoir example in Part 2.
- § Direcção-Geral do Território. Portuguese national height datum, which fixes zero for land heights at mean sea level measured at Cascais; the basis of the sea level discussion in Part 2.
- § British Antarctic Survey. Emperor penguin breeding biology and typical winter air temperatures at colony sites, used for the penguin figure in Part 2.
- § Bureau International des Poids et Mesures. The International System of Units, ninth edition, for the definition and correct written form of the degree Celsius.
- § Prime School Mathematics Department. Year 4 scheme of work, 150 hours, terms 1 to 3, which fixes the unit and topic order followed by this book.
- § Prime School weather station log, Setúbal campus, January readings, from which the shape of the week's data in Part 2 was drawn.

**On the illustrations.** All artwork in this book was created for this edition. Diagrams, number lines, place value charts, thermometers, sequence chains and tables are drawn as vector artwork by the Prime Books typesetting system from the numbers they represent, which is why a figure in this book can never disagree with its own caption.

**On the numbers.** Every sequence, decomposition, temperature difference, ordering and marking total in this book was generated and verified by computer during production. The answer section is derived from the same source as the questions rather than typed separately.

# Both Ways from Zero

Numbers do not stop at zero. This year, Year 4 follows the line in both directions: up into the thousands, and down past zero into the cold.

Unit 1 begins with a real question. On one January morning the thermometer at the Torre read  $-6^{\circ}\text{C}$  while Faro read  $13^{\circ}\text{C}$ . How far apart are they? Answering that properly takes counting, a number line, and a clear head about which side of zero you are on.

- ✓ Every topic opens with a worked example that shows its answer
- ✓ A real number line wherever a number line is mentioned
- ✓ A full place value chart, with what every digit is worth
- ✓ Over sixty practice questions, and twelve fresh ones to check yourself
- ✓ Watch out! panels that name the mistake before you make it
- ✓ A review page that says confident, almost there or needs more work

Written by the Prime School Mathematics Department for the school's Year 4 pupils, and reviewed in manuscript by the teachers who teach it.